

Pramo Jewels

Payment Gateway Integration Specification

Document ID: PJ-PGI-001

Version: 1.0

Status: Approved for Development

Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define payment gateway architecture, workflows, security controls and integrations
Audience	Backend Developers, DevOps, QA, Finance Team, Security Team
Version	1.0

Supported Payment Features

Payment Method	Specification
UPI	Collect, authorize and verify UPI payments via gateway APIs
Cash on Delivery	Configurable by location, order value and product category
Future Methods	Cards, Net Banking, Wallets and EMI (future-ready)
Refunds	Full and partial refund support
Webhooks	Signed webhook verification and event processing
Reconciliation	Daily reconciliation with orders and finance
Fraud Controls	Risk checks, idempotency and transaction monitoring

Architecture

The payment gateway acts as an external service integrated through secure REST APIs. The platform never stores raw card credentials and relies on gateway tokenization where applicable.

Payment Flow

Checkout → Create Payment Order → Customer Pays (UPI/COD) → Gateway Response → Webhook Verification → Order Confirmation → Invoice Generation.

Business Rules

Verify payment before fulfillment except for COD. Use idempotency keys to prevent duplicate transactions. Handle timeout and retry scenarios gracefully.

Webhook Processing

Validate digital signatures, process events asynchronously, ensure replay protection and maintain event audit logs.

Security

Enforce HTTPS/TLS, encrypt sensitive data, protect API keys in secret management, apply rate limiting and follow PCI DSS principles.

Failure Handling

Support retries for transient failures, reconciliation for uncertain states and customer-friendly error messages without exposing sensitive details.

Integrations

Integrates with Cart, Checkout, Orders, Inventory, Notifications, Accounting, Analytics and Fraud Monitoring.

Monitoring

Track payment success rate, failure rate, latency, webhook health, refund processing time and reconciliation status.

Acceptance Criteria

Payments are processed securely, duplicate transactions are prevented, reconciliation is accurate and all payment events are auditable.